PRINCE CHAUHAN

Senior Thermal Engineer · Battery & charger thermal management

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Scan: the work in 3D

SUMMARY

Thermal engineer with 3+ years designing liquid- and air-cooled thermal management for battery-swap and fast-charging stations from 20 to 110 kWh, holding packs inside a 25–35 °C window at ambient temperatures up to 55 °C: first-principles heat-load estimation, coolant-loop sizing, radiator and chiller selection against ΔT budgets, chamber and field validation, and the control logic that runs it. Four patent applications filed. Working towards intelligent, data-driven thermal systems and spacecraft thermal control.

EXPERIENCE

SENIOR THERMAL ENGINEER · SUN MOBILITY

2023 – present

Electric Vehicles Dept · Bengaluru

SUPERCHARGE ARCHITECTURE VIEW IN 3D · POC

- Architected and validated the coolant system for a supercharge demonstrator: **650 kg packs held at 25–35 °C** through 110 kWh supercharging at **60 °C ambient, with no thermal fault**.
- Led thermal validation at OEM factories: test-setup design, data-acquisition protocol, reporting, and training factory engineers on the procedure.

LIQUID-COOLED CHARGER & BATTERY VIEW IN 3D · PROTOTYPE

- Defined the BTMS/CTMS thermal architecture for 20–110 kWh platforms to 55 °C ambient: heat-load estimation by energy balance and loss modelling, coolant-flow sizing, radiator and chiller selection against ΔT limits.
- Qualified supplied hardware against datasheet in the chamber, finding a chiller short of its rating as ambient rose, and carried the shortfall into the sizing margin.
- Worked with battery manufacturers on iterative cooling design, cutting charging thermal resistance by **18%**; optimised radiator sizing and flow distribution under peak charging loads (1.6C–3C).
- Designed a redundant-pump liquid loop for the charger rectifier stage; testing showed derating above 50 °C beyond datasheet, fed back to the supplier.

INTEGRATED BTMS + CTMS · PILOT VIEW IN 3D · PILOT

- Brought the BTMS and CTMS into a single enclosed, serviceable unit on the 110 kWh series-ready pilot rack.
- Authored the System of Requirements (SOR) defining BTMS/CTMS state machines, flow diagrams and alert logic, adopted as the governing control logic by the embedded team.
- Built an ECU–station controller simulator and over-charge cooling safety logic to stress-test the control strategy across the temperature envelope **before hardware**.

AIR-COOLED CHARGERS

- Air-cooled charging station, taken from proof of concept to station design: **Best Team Award** (Air-cool station & POC team, Q3 FY2025–26).

SIMULATION ENGINEER · GTU MOTORSPORTS

2018 – 2022

Formula Bharat · Gujarat

- Led chassis optimisation with CAD and FEA, cutting vehicle weight **15%** and contributing to a national win against 80+ teams including IITs and NITs (2019).

PATENTS & RECOGNITION

- **IN 202441065103** · Thermal management of chargers · filed
- **IN 202441065106** · Thermal management of energy storage devices · filed
- Two further applications filed (2025): dynamic compressor-speed control for battery thermal systems, and reinforcement-learning-based liquid cooling control.
- **InvenZone Award 2025** (IP contribution) · **Best Team Award**, SUN Mobility (Q3 FY2025–26)

SKILLS

THERMAL DESIGN	BTMS · CTMS · heat-load estimation · coolant-loop sizing · radiator/chiller selection · thermal-runaway prevention · multi-scale modelling (cell / pack / station)
SIMULATION	ANSYS Fluent (CFD) · Siemens NX Space Systems Thermal · Flow Simulation · SolidWorks · PTC Creo · FEA
TEST & VALIDATION	Chamber testing (–10 to 60 °C) · RTD / thermocouple calibration · IR thermography · DAQ · DOE · RCA · supplier qualification
CONTROLS	Thermal state machines · ECU simulation · alert & safety logic · CAN 2.0 · real-time monitoring

SPACECRAFT THERMAL

- Self-directed satellite thermal-control study (hot-case radiator sizing, eclipse cold-case heater sizing, optical-bench gradients) and an eight-part public series on satellite thermal architecture.

EDUCATION & CERTIFICATIONS

DEGREE	2022 · B.E., Automobile Engineering · LJIET, Gujarat · project: 3 kW solar tree (~50% cost, ~80% footprint vs a fixed array)
CERTIFICATIONS	MITx Astronautics & Human Spaceflight · DelftX EV Technology · DelftX Solar Energy · Coursera ML · IBM Data Science